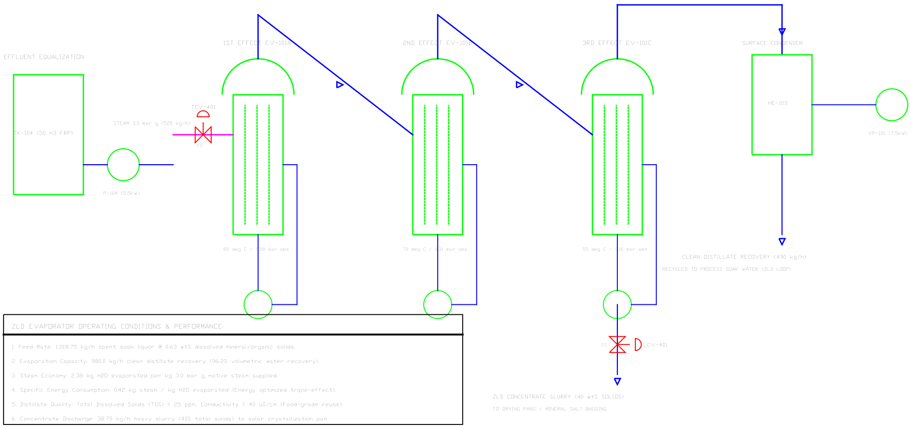
EFFLUENT EQUALIZATION
TK-104 (50 m3 FRP)
P-104 (5.5kW)
STEAM 3.0 bar g (520 kg/h)
TCV-401
FC
1ST EFFECT EV-101A
85 deg C / 0.58 bar abs
2ND EFFECT EV-101B
70 deg C / 0.31 bar abs
3RD EFFECT EV-101C
55 deg C / 0.16 bar abs
SURFACE CONDENSER
HE-103
VP-101 (7.5kW)
CLEAN DISTILLATE RECOVERY (490 kg/h)
RECYCLED TO PROCESS SOAK WATER (ZLD LOOP)
FC
LCV-401
ZLD CONCENTRATE SLURRY (40 wt% SOLIDS)
TO DRYING PANS / MINERAL SALT BAGGING

ZLD EVAPORATOR OPERATING CONDITIONS & PERFORMANCE:

1. Feed Rate: 1,018.75 kg/h spent soak liquor @ 0.63 wt% dissolved mineral/organic solids.
2. Evaporation Capacity: 980.0 kg/h clean distillate recovery (96.2% volumetric water recovery).
3. Steam Economy: 2.38 kg H2O evaporated per kg 3.0 bar g motive steam supplied.
4. Specific Energy Consumption: 0.42 kg steam / kg H2O evaporated (Energy optimized triple-effect).
5. Distillate Quality: Total Dissolved Solids (TDS) < 25 ppm, Conductivity < 40 uS/cm (Food-grade reuse).
6. Concentrate Discharge: 38.75 kg/h heavy slurry (40% total solids) to solar crystallization pan.

CONCEPTUAL FEED - PRELIMINARY - NOT FOR CONSTRUCTION / NOT FOR PROCUREMENT

HARAKUTIR AGRO-PROCESS ENGINEERING PVT. LTD.
15,000 kg/day Split Pigeon Pea Optimization Greenfield Project

P&ID - TRIPLE-EFFECT ZLD WASTEWATER EVAPORATOR
DISCIPLINE: PIPING | STAGE: CONCEPTUAL FEED

DWG NO: DWG-FIN-15T-010
REV: V4.0
SHEET: 10 OF 12
SCALE: AS SHOWN
DATE: 2026-10-15
STATUS: PRELIMINARY
GATE: G-A HOLD